

An Efficient Multi-Frame Track-Before-Detect Algorithm for Multi-Target Tracking

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Abstract—This paper considers the multi-target tracking (MTT) problem through the use of dynamic programming based track-before-detect (DP-TBD) methods. The usual solution of this problem is to adopt a multi-target state, which is the concatenation of individual target states, then search the estimate in the expanded multi-target state space. However, this solution involves a high-dimensional joint maximization which is computationally intractable for most realistic problems. Additionally, the dimension of the multi-target state has to be determined before implementing the DP search. This is problematic when the number of targets is unknown. We make two contributions towards addressing these problems. Firstly, by factorizing the joint posterior density using the structure of MTT, an efficient DP-TBD algorithm is developed to approximately solve the joint maximization in a fast but accurate manner. Secondly, we propose a novel detection procedure such that the dimension of the multi-target state no longer needs to be pre-determined before the DP search. Our analysis indicates that the proposed algorithm could achieve a computational complexity which is almost linear to the number of processed frames and independent of the number of targets. Simulation results show that this algorithm can accurately estimate the number of targets and reliably track multiple targets even when targets are in proximity.

Index Terms—Dynamic programming, high-dimensional estimation, multiple target tracking, track-before-detect.

I. INTRODUCTION

THE classical tracking approaches [1], such as probabilistic data association (PDA) and multiple hypothesis testing (MHT) all use thresholded detections as input. One weakness of these approaches is that much of the information contained in the measurements is irreversibly discarded after thresholding. This information reduction is harmful to the detection of the low signal-to-noise ratio (SNR) targets. Compared to conventional tracking methods, track-before-detect (TBD) can achieve superior performance for tracking of low SNR targets by jointly processing several consecutive frames of

unthresholded data, or thresholded data with significantly lower thresholds than are used by conventional trackers. Its efficiency and robustness have been widely demonstrated [2]–[7]. In addition, TBD avoids the need for direct measurement-to-track association, since the association is embedded implicitly in TBD framework.

Existing approaches to TBD include dynamic programming (DP) [8], [9] algorithms, particle filter (PF) based methods [10]–[13] and the multi-Bernoulli random set filter [14]. In this paper, we focus on the DP based TBD (DP-TBD) methods. By discretizing the state space, DP-TBD is able to implement Bayesian filtering and approximate the posterior density by jointly processing multiple consecutive data frames. Since all measurements are processed simultaneously, DP-TBD is a natural method to monitor the entire surveillance area and detect newborn targets. A drawback of DP-TBD is its potentially high computational complexity because it involves processing a large volume of data. Therefore, an efficient implementation is the key issue for DP-TBD. This is particularly true when tracking multiple targets.

The DP-TBD method was first introduced in [15], [16], and has been well studied in recent years [5]–[7], [17]. Its application has been extended to other areas such as radar [18]–[22] and sonar [23], [24] systems as well. However, all these works assumed that only a single target appears in the surveillance area, which we refer to as single-target DP-TBD (ST-DP-TBD), while the multiple target tracking (MTT) problem is generally more important in real-world applications.

Similar to other multi-target TBD algorithms, such as [10]–[12], an immediate extension of DP-TBD to the MTT problems can be realized by adopting a multi-target state, which is the concatenation of several individual target states. Track estimates are then sought in the expanded state space. Although the implementation of the multi-target state based DP-TBD is straightforward in principle, it has two major drawbacks. Firstly it involves the solution of a high-dimensional maximization which is computationally prohibitive if there are more than two targets. Secondly, it does not easily accommodate the important practical case of an unknown number of targets.

In [5], the authors provided a brief and intuitive way of multi-target tracking using DP-TBD, but did not address the problem in real detail. Later in [26], an efficient multi-path Viterbi-like algorithm [27]–[29] was employed to solve the high-dimensional optimization problem, but it still has a complexity which is exponential in the number of targets, so it is not viable for even a moderate number of targets. To reduce the computational complexity, in [26], two sub-optimal algorithms were also developed by approximation of the optimal multi-path DP search. Although the light computational expense makes them the first im-

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plementable DP-TBD solutions to the practical MTT problem, their tracking performance may suffer seriously due to the interference when targets are in close proximity. In addition, a multiple hypothesis testing method is employed to accommodate the case of an unknown number of targets. An upper bound on the number of prospective targets $M_{\max}$ needs to be pre-defined. Generally, $M_{\max}$ should be large enough to cover the possibility of the largest number of targets, but a big $M_{\max}$ causes unnecessary computational expense and performance loss due to the increased number of admissible hypotheses.

This paper is an extension of our preliminary work [30]. It considers the tracking of an unknown number of targets through the use of DP-TBD method, and makes two contributions as follows towards addressing the limitations of existing algorithms described in the preceding paragraphs.

1) *The Structure of MTT Problem*: the high dimensional maximization problem in DP-TBD is solved sub-optimally in a simpler and cheaper way by utilizing the structure of MTT problem [25]. When targets are clustered into well separated groups, there is no measurement ambiguity between distant targets and therefore the target groups can be treated independently. This idea has already been used in particle filter based MTT methods [11]–[13] to enhance the sampling efficiency, but it is an unexplored area for DP-TBD methods. By incorporating this idea into DP integration, we show that the high-dimensional maximization can be broken down to several maximizations of lower dimension, thus the computational cost is significantly reduced. On the other hand, targets in the same cluster are still considered jointly, so the tracking performance is resilient to target interference.

2) *Generalized Detection Procedure*: all the aforementioned DP-TBD algorithms use the same detection procedure: after DP integration, the target state with the global maximum merit is selected and tested against a detection threshold. If the global maximum merit exceeds the threshold, the estimated (single-target or multi-target state) trajectory is declared and recovered, otherwise the null hypothesis of no target is accepted. In this paper, motivated by our analysis of the single-target DP merit function, a generalized detection procedure is developed. A key advantage of this detection procedure is that it requires only a single-target state based DP integration without regard for how many targets exist. After the single-target DP integration, not only the global maximum, but every merit is tested against the detection threshold. All the exceeded elements are retained to form an estimation candidate group from which the partitions of the estimated multi-target state, i.e., individual target states, will be extracted recursively in the subsequent processing over relatively small subsets of the state space. The estimated number of targets is then the number of the extracted individual target states.

The rest of the paper is organized in the following manner. In Section II the target dynamic model and measurement model along with some notation are given. The background and some important concepts of multi-target DP-TBD are presented in Section III. In Section IV, we analyze the single target DP merit

function. This motivates the development of the proposed algorithm in Section V. Simulation results examining the performance of the proposed algorithm, including comparisons with the algorithms in [26], are presented in Section VI. Finally, conclusions are drawn in Section VII.

II. MODELS AND NOTATION

A. Target Dynamic Model

Let K denote the number of consecutive frames processed in a DP integration batch. Consider that a fixed and unknown number M of point targets move independently in a two-dimensional (2-D) $x - y$ plane. Since K is typically quite small in DP-TBD the assumption of a constant M is not restrictive. The motion of individual targets can be modeled in state space $\mathbb{R}^s$, $s = 4$. The state of the m th target, $m = 1, \dots, M$, at stage k is $\mathbf{x}_k^m = [p_{x,k}^m, v_{x,k}^m, p_{y,k}^m, v_{y,k}^m]'$, where $'$ denotes the matrix transpose, $p_{x,k}^m, p_{y,k}^m$ denote the target positions, and $v_{x,k}^m, v_{y,k}^m$ denote the velocities in the x and y dimensions respectively. It should be pointed out that although the dimension of measurement space and state space are 2-D and 4-D here, the extension to a higher dimensional case is direct. The multi-target state at time k is defined as the concatenation of the individual target states, $\mathbf{X}_k = [(\mathbf{x}_k^1)'; (\mathbf{x}_k^2)'; \dots; (\mathbf{x}_k^M)']'$. The state evolution of individual targets satisfies the linear Markov model, and the process noise is Gaussian distributed, thus the state of the m th target evolves according to

$$\mathbf{x}_k^m | \mathbf{x}_{k-1}^m \sim \mathcal{N}(\cdot; \mathbf{F}\mathbf{x}_{k-1}^m, \mathbf{Q}), \quad (1)$$

where $\mathcal{N}(\mathbf{x}; \boldsymbol{\mu}, \boldsymbol{\Sigma})$ denotes the Gaussian probability density evaluated at $\mathbf{x}$ with mean $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$, and $\mathbf{F}$ and $\mathbf{Q}$ are the state transition matrix and the process noise covariance matrix given by

$$\mathbf{F} = \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix} \otimes \mathbf{I}_2, \quad (2)$$

$$\mathbf{Q} = \kappa \begin{bmatrix} T^3/3 & T^2/2 \\ T^2/2 & T \end{bmatrix} \otimes \mathbf{I}_2, \quad (3)$$

where T denotes the time interval between successive frames, κ is the process noise intensity, $\mathbf{I}_l$ denotes the $l \times l$ identity matrix and $\otimes$ is the Kronecker product. The proposed algorithm is not restricted to the use of the model (1). For example, in [26] it is assumed that the state at time k is uniformly distributed over a set of allowable states. The proposed algorithm could also be applied with this model.

The trajectory of the m th target is a sequence of successive states from the 1st frame to the K th frame and is denoted as $\mathbf{X}_{1:K}^m = (\mathbf{x}_1^m, \mathbf{x}_2^m, \dots, \mathbf{x}_K^m)$. Similarly, the trajectories of all M targets are defined as a sequence of successive multi-target states, $\mathbf{X}_{1:K} = (\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_K)$.

B. Measurement Model

Generally, TBD methods are considered on the pixelized model which has often been used to depict low resolution

sensors [2]. The surveillance region is divided into an $N_x \times N_y$ grid of cells, where N_x and N_y are the number of cells in x and y axis respectively. The k th measurement frame $\mathbf{z}_k$ is defined as a $N_x \times N_y$ array whose (i, j) th element $z_k^{i,j} \in \mathbb{R}$, $i = 1, \dots, N_x$, $j = 1, \dots, N_y$, is the measurement intensity recorded in the (i, j) th resolution cell. The K frames in a DP processing batch are expressed as $\mathbf{Z}_{1:K} = (\mathbf{z}_1, \dots, \mathbf{z}_K)$.

Measurements are assumed to be independently distributed in each cell with the distribution depending on the number of targets in this cell. Let $m^{i,j}(\mathbf{X}_k)$ denote the number of targets in cell (i, j) with state $\mathbf{X}_k$. Generally, it is expected that $z_k^{i,j}$ is likely to have a larger value as $m^{i,j}(\mathbf{X}_k)$ increases. Note that a target affects only the measurement of the cell which it occupies. Rayleigh distributed measurements are employed:

$$\ell(z_k^{i,j}|\mathbf{x}_k) = \frac{z_k^{i,j}}{1 + \vartheta m^{i,j}(\mathbf{X}_k)} \exp\left(\frac{-(z_k^{i,j})^2}{2(1 + \vartheta m^{i,j}(\mathbf{X}_k))}\right), \quad (4)$$

where ϑ denotes the target SNR, assumed to be known. Similar models have been used in multi-target tracking [11], [12], and monopulse radar applications [31]. Note that the development of DP-TBD algorithm does not require a particular form of $\ell(\cdot|\mathbf{X}_k)$. Further, when $\ell(\cdot|\mathbf{X}_k)$ is unknown, the measurement amplitude can be approximately used as integration merit [6].

III. BACKGROUND AND PROBLEM STATEMENT

The aim is to estimate both the number of targets present and their states. From a Bayesian perspective, $\hat{\mathbf{X}}_{1:K} = (\hat{\mathbf{X}}_1, \dots, \hat{\mathbf{X}}_K)$, i.e., the estimated target state sequence, which implicitly includes an estimate of the number of targets, can be found from the posterior density function $p(\mathbf{X}_{1:K}|\mathbf{Z}_{1:K})$ given a certain estimation criterion. However, the analytic solution for calculating $p(\mathbf{X}_{1:K}|\mathbf{Z}_{1:K})$ is usually intractable. By discretizing the continuous state space $\mathbb{R}^s$ into a multidimensional grid with a finite number of cells, DP-TBD is able to implement Bayesian filtering in the discretized state space $\overline{\mathbb{R}}^s \subset \mathbb{R}^s$ and then approximate the posterior density. The estimated state sequence $\hat{\mathbf{X}}_{1:K} \in \overline{\mathbb{R}}^{sMK}$ is the one which maximizes $p(\mathbf{X}_{1:K}|\mathbf{Z}_{1:K})$ in the discrete state space $\overline{\mathbb{R}}^{sMK}$, i.e., the maximum a posterior probability (MAP) estimate. Generally, the position dimensions of $\overline{\mathbb{R}}^s$ are discretized by the unit size Δ of the measurement frame, and the velocity dimensions are divided by $\dot{\Delta}$, i.e., $1 \Delta/\text{frame}$. Thus the discrete state space $\overline{\mathbb{R}}^s$ is a 4-dimensional grid of $N_x \times (2V^{\max} + 1) \times N_y \times (2V^{\max} + 1)$ cells, where $V^{\max}$ denotes the maximum number of cells a target can travel during a frame interval. For a continuous target state $\mathbf{x}_k = (p_{x,k}, v_{x,k}, p_{y,k}, v_{y,k})'$, its corresponding discrete state is defined as $(\lceil p_{x,k}/\Delta \rceil, \lceil v_{x,k}/\dot{\Delta} \rceil, \lceil p_{y,k}/\Delta \rceil, \lceil v_{y,k}/\dot{\Delta} \rceil)'$.

A. Single-Target DP-TBD

If only a single target appears, (i.e., $M = 1$), ST-DP-TBD finds a most probable single-target state sequence as

$$\begin{aligned} \hat{\mathbf{X}}_{1:K}^1 &= \arg \max_{\mathbf{X}_{1:K}^1 \in \overline{\mathbb{R}}^{sK}} p(\mathbf{X}_{1:K}^1|\mathbf{Z}_{1:K}) \\ \text{s.t. } p(\hat{\mathbf{X}}_{1:K}^1|\mathbf{Z}_{1:K}) &> V_{DT}, \end{aligned} \quad (5)$$

where V_{DT} is a detection threshold determined by the detection or false alarm probabilities. If $p(\hat{\mathbf{X}}_{1:K}^1|\mathbf{Z}_{1:K})$ exceeds V_{DT} , a target is declared, otherwise null is declared. Using Bayes' rule, the posterior density can be expanded as,

$$p(\mathbf{X}_{1:K}^1|\mathbf{Z}_{1:K}) \propto \prod_{k=1}^K p(\mathbf{z}_k|\mathbf{x}_k^1) p(\mathbf{x}_k^1|\mathbf{x}_{k-1}^1). \quad (6)$$

Using (6), the computation of (5) can be interpreted as finding the best path in a trellis diagram of length K whose states are the elements of space $\overline{\mathbb{R}}^s$. The DP algorithm [8], [9] provides a solution to the maximization in (5) by efficiently enumerating and evaluating all possible state sequences in space $\overline{\mathbb{R}}^{sK}$.

For $2 \leq k \leq K$, and for all $\mathbf{x}_k^1 \in \overline{\mathbb{R}}^s$

$$\begin{aligned} I(\mathbf{x}_k^1|\mathbf{Z}_{1:k}) &= \max_{\mathbf{x}_{k-1}^1 \in \tau(\mathbf{x}_k^1)} [I(\mathbf{x}_{k-1}^1|\mathbf{Z}_{1:k-1}) + Tr(\mathbf{x}_k^1|\mathbf{x}_{k-1}^1)] \\ &\quad + L(\mathbf{z}_k|\mathbf{x}_k^1) \end{aligned} \quad (7)$$

where $I(\mathbf{x}_k|\mathbf{Z}_{1:k})$ is the merit function of state $\mathbf{x}_k$, and consists of three parts: $L(\mathbf{z}_k|\mathbf{x}_k^1)$, $Tr(\mathbf{x}_k^1|\mathbf{x}_{k-1}^1)$ and $I(\mathbf{x}_{k-1}^1|\mathbf{Z}_{1:k-1})$. The log likelihood $L(\mathbf{z}_k|\mathbf{x}_k^1) = \log\left(\prod_{i=1}^{N_x} \prod_{j=1}^{N_y} \ell(z_k^{i,j}|\mathbf{x}_k^1)\right)$ represents the contribution of the measurements associated to the state $\mathbf{x}_k^1$. The term $Tr(\mathbf{x}_k^1|\mathbf{x}_{k-1}^1)$, referred to as the state transition cost function, assigns a penalty corresponding to the probability of transition between state $\mathbf{x}_{k-1}^1$ and $\mathbf{x}_k^1$. The maximization is performed over $\tau(\mathbf{x}_k^1)$ which is a collection of discrete target states at stage $k-1$ from which a transition to $\mathbf{x}_k^1$ is possible. After DP integration, a detection is made by testing the global maximum merit with V_{DT} . The complexity of ST-DP-TBD is about $\mathcal{O}(|\overline{\mathbb{R}}^s|K|\tau|)$, where $|\overline{\mathbb{R}}^s|$ and $|\tau|$ denote the number of the elements in space $\overline{\mathbb{R}}^s$ and in the state collection $\tau(\mathbf{x}_k^1)$ respectively.

B. Multi-Target State Based DP-TBD and its Sub-Optimal Approximations

Similarly to other multi-target algorithms such as particle filters [11]–[13], the multi-target state based DP-TBD methods are a direct extension of ST-DP-TBD to multi-target scenarios. In a scenario with $M > 1$ targets, if M is known, the track estimate of the multi-target state based DP-TBD is defined as

$$\begin{aligned} \hat{\mathbf{X}}_{1:K} &= \arg \max_{\mathbf{X}_{1:K} \in \overline{\mathbb{R}}^{sMK}} p(\mathbf{X}_{1:K}|\mathbf{Z}_{1:K}), \\ \text{s.t. } p(\hat{\mathbf{X}}_{1:K}|\mathbf{Z}_{1:K}) &> V_{DT}. \end{aligned} \quad (8)$$

If $p(\hat{\mathbf{X}}_{1:K}|\mathbf{Z}_{1:K})$ exceeds V_{DT} , a detection of M targets is declared, otherwise no target is declared. Similarly, using Bayes' rule as in (6), solving (8) can be viewed as finding M physically-admissible paths, which maximize the posterior density function, in the expanded trellis diagram of length K .

Although (8) can be solved straightforwardly in principle, this procedure is computationally prohibitive if there are more than a few targets. Besides, the dimension of the multi-target state has to be pre-defined before carrying out DP search. If M is unknown, all possible hypotheses of M have to be considered (i.e., a multiple hypotheses testing problem), which makes (8) even more expensive and difficult. Hence, the predominant challenge

is to find a computationally efficient solution, or rather approximate solution, for (8).

In [26], the multi-path DP algorithm [27]–[29] is employed to solve (8) with a complexity $\mathcal{O}((|\mathbb{R}^s|M)^2 |\tau| M^3 K)$. To reduce the computational complexity, two suboptimal solutions (i.e., STC-VTA and SP-STC-VTA) were also developed to approximately implement the optimal multi-path DP search. However, the performance of these suboptimal approaches degrades when targets are near each other, especially SP-STC-VTA. Moreover, for the case of an unknown number of targets, a multiple hypothesis testing algorithm is employed as the detection procedure. At each detection time, the merit function weights for all the $M_{\max}$ (the assumed upper bound on the number of targets) possible hypotheses have to be computed before a decision is made. Consequently, the resulted complexity of STC-VTA and SP-STC-VTA are about $\mathcal{O}(|\mathbb{R}^s|M_{\max}K|\tau|)$ and $\mathcal{O}(|\mathbb{R}^s|K|\tau|)$ respectively. Since the actual number M of targets may be far less than $M_{\max}$, this detection procedure is not computationally efficient.

IV. SINGLE TARGET DP MERIT FUNCTION

The joint maximization in (8) implies that the multi-target state based DP-TBD method considers all existing targets jointly. However, the structure of MTT problem [25] suggests that only closely spaced target needs to be considered jointly, while well separated targets can be considered independently. If the M targets in a scenario can be divided into $l \leq M$ well separated clusters, the joint likelihood satisfies [12],

$$p(\mathbf{z}_k | \mathbf{X}_k) \approx \alpha \prod_{i=1}^l \pi(\mathbf{z}_k | \mathbf{C}_k^i), \quad (9)$$

where $\mathbf{C}_k^i$ is the multi-target state of the i th cluster at the k th frame. The term α is a constant independent of $\mathbf{X}_k$, and the notation $\pi(\cdot | \mathbf{C}_k^i)$ denotes the likelihood of the measurement conditional on the i th target cluster under the assumption that only this target cluster exists. $\pi(\cdot)$ will be used throughout this paper to indicate densities evaluated in this manner.

This likelihood factorization (9) has been utilized in the multi-target state based particle filters [11]–[13] to enhance the sampling efficiency, i.e., the particles are drawn jointly for the targets in the same cluster, but independently between clusters. Regarding to DP-TBD, we use (9) to split the high-dimensional maximization in (8) into several low-dimensional maximizations as

$$\begin{aligned} p(\mathbf{X}_{1:K} | \mathbf{Z}_{1:K}) &\approx \prod_{i=1}^l \pi(\mathbf{C}_{1:K}^i | \mathbf{Z}_{1:K}) \\ &\propto \prod_{i=1}^l \prod_{k=1}^K \pi(\mathbf{z}_k | \mathbf{C}_k^i) p(\mathbf{C}_k^i | \mathbf{C}_{k-1}^i). \end{aligned} \quad (10)$$

Intuitively, for a special scenario where all targets are far away from each other, the joint maximization (4KM-D) of (8) is broken down into M maximizations of dimension $4K$, which means the complexity is linear in the number of targets.

In order to integrate (10) into the DP-TBD implementation, here the single-target DP merit function is analyzed. Figs. 1 and 2 show the final stage merit function for two typical tracking

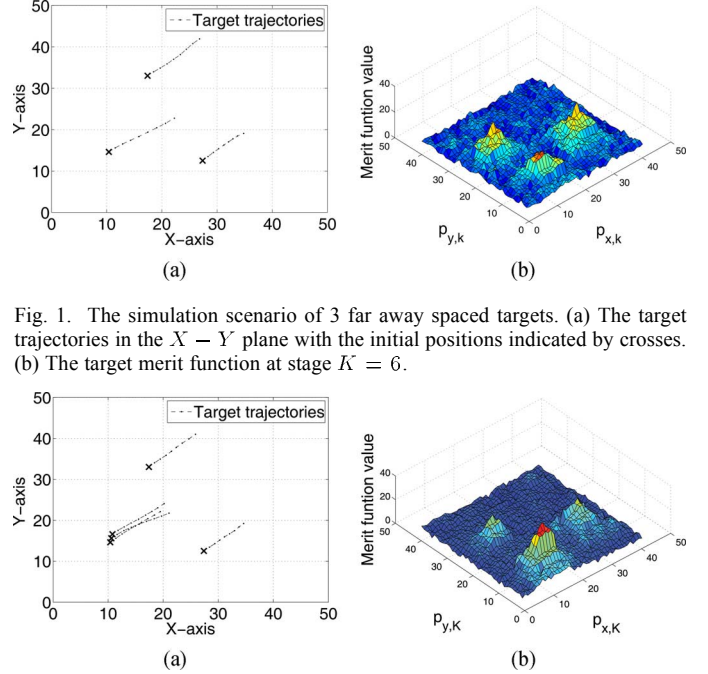


Fig. 1. The simulation scenario of 3 far away spaced targets. (a) The target trajectories in the $X-Y$ plane with the initial positions indicated by crosses. (b) The target merit function at stage $K = 6$.

Fig. 2. The simulation scenario of 5 targets, where three targets are in proximity. (a) The target trajectories in the $X-Y$ plane with the initial positions indicated by crosses. (b) The target merit function at stage $K = 6$.

scenarios.¹Fig. 1(a) first shows a scenario with 3 well-separated targets. Fig. 1(b) is the corresponding merit function at stage $K = 6$ after implementing a ST-DP-TBD. It can be seen clearly that there are three separated merit peaks, each peak corresponds to a target sequence. Thus, if we can extract the states of the three merit peaks, three targets can be detected simply by implementing a ST-DP-TBD. We refer to this situation as “targets are resolved in the merit function.”

Fig. 2 is a more difficult scenario: five targets, with three in proximity. It can be seen that the merit functions of the three adjacent targets merge together. This is because the DP enumerates all the possible target states during integration. If target trajectories are close to each other, the DP maximization function will merge them and form a single merit peak. We refer to this situation as “targets are unresolved in the merit function.” To solve this problem, the multi-target state based DP-TBD adopts a multi-target state with five individual target states to consider all the five targets jointly. Although this is theoretically optimal in terms of detection, it is computationally inefficient and does not take account into the structure of MTT problem. Based on the factorization in (10), only the three adjacent targets need to be considered jointly, the two far away targets can be treated independently to reduce the complexity. The proposed algorithm is developed in the following section to achieve this goal.

Another issue worth mentioning is that all the merit peaks are expanded, i.e., “merit function expansion phenomenon” [17]. After the DP integration, the merit function of the true target state is considerably increased, usually becoming the largest one. Meanwhile the merit functions of the states which are close to the true state are also increased a lot by sharing one or more possible previous states with the true target trajectory.

¹In order to visualize the 4-D merit function in 2-D figures, the velocity dimensions of target states are fixed at the real target velocities.

The closer to the true target trajectory, the higher the merits of the states will be. Thus a lobe-shape merit function centered at the true target state is formed. The shape of the merit function expansion is determined by the size of the possible transition area $\tau(\mathbf{x}_k^1)$, the target signal intensity and the length of integration period. We can see that the three peaks in Fig. 1(b) have similar shapes, because the integration parameters are the same. In Fig. 2(b), the joint peak for the three adjacent targets is broader and higher, this is because the measurements of the three targets all contribute to this joint peak. This implies that the shape of the merit function may reflect the number of targets in this peak and their relative positions. This paper only exploits the structure of merit function to break down the high-dimensional optimization problem of multi-target tracking. The exploitation of the shape of merit function to group target tracking and extended target tracking may also be interesting, though rigorous models have to be established.

V. DEVELOPMENT OF THE TRACKING ALGORITHM

Based on the merit function analysis in the preceding section, a multi-target DP-TBD algorithm (MT-DP-TBD) is developed to take account into the structure of MTT problem. Besides, the proposed algorithm adopts a generalized detection procedure to deal with an unknown number of targets instead of using a multi-hypotheses testing algorithm.

To do that, MT-DP-TBD is composed of three sequential processes: integration and thresholding (IAT), sorting and merging (SAM) and parallel target cancellation DP (PTC-DP). In brief, the IAT process first implements a single-target DP integration, then applies a detection threshold to the merit function. All target states whose merit functions exceed the threshold are stored in a collection of state candidates. This collection contains both real target states and false target states which caused by noise and the merit function expansion phenomenon. The purpose of SAM is then to eliminate the redundant and inappropriate state candidates, and extract real target states from this collection. In the special cases where targets are all well separated, multiple targets can be detected via the combination of just IAT and SAM. However, if there are closely spaced targets, the merit peaks of the nearby targets become “unresolved in the merit function” as shown in Fig. 2(b). The PTC-DP process is designed to uncover these adjacent targets by repeatedly searching the potential target-hidden areas after removing the interference of the former detected targets. How to locate these target-hidden areas and how to select the size of these areas are two important issues for PTC-DP. We will discuss this in Section V-C.

In Section V-A, the special scenario wherein targets are all well separated is considered first to develop IAT and SAM. The more interesting case, tracking arbitrarily located targets, will be addressed when developing PTC-DP in Section V-B.

A. Tracking Well Separated Targets

Fig. 3 is a sketch map of a scenario with M well separated targets. It can be seen that the M trajectories are contained in M separated circular areas $S^1, \dots, S^M$. The m th circular area $S^m \subset \mathbb{R}^{s/2}$ has radius $R_{\max}$, centered at the initial position of the m th target trajectory $[p_{x,1}^m, p_{y,1}^m]$. The term

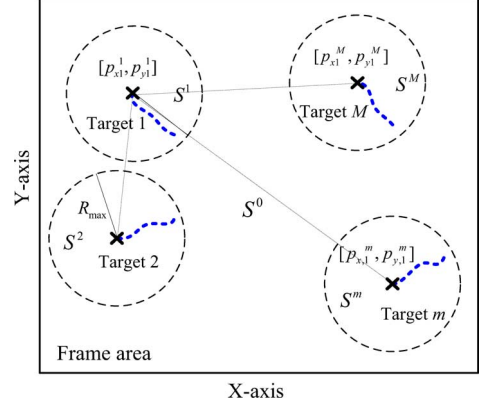


Fig. 3. Sketch map of a scenario wherein all M targets are well separated. The dash circles indicate the circular areas $S^1, \dots, S^M$ containing the target trajectory. The remaining part of the scenario S^0 only contains the noise background. The dash lines denotes target trajectories with initial position indicated by the crosses.

$R_{\max} = KT\sqrt{(V_x^{\max})^2 + (V_y^{\max})^2}$ is the maximum distance a target can travel during the interval of K frames. Let $\bar{\mathbb{R}}^s(S^m)$ denotes the subsets of the discretized state space corresponding to the m th circular area S^m , and $\mathbf{Z}_{1:K}(S^m)$ denotes the measurement sequence taken in area S^m . It can be seen that only the m th target contributes to the measurements $\mathbf{Z}_{1:K}(S^m)$. The remaining measurements $\mathbf{Z}_{1:K}(S^0)$ contain only noise measurements.

For this kind of scenarios, the posterior density satisfies

$$p(\mathbf{X}_{1:K}|\mathbf{Z}_{1:K}) \propto \prod_{m=1}^M \pi(\mathbf{X}_{1:K}^m|\mathbf{Z}_{1:K}(S^m)), \quad (11)$$

where $\propto$ means “is approximately proportional to.” Thus the estimation of the individual target state in each component of (11) can be treated as a ST-DP-TBD problem as

$$\hat{\mathbf{X}}_{1:K}^m = \arg \max_{\mathbf{X}_{1:K}^m \in \bar{\mathbb{R}}^{s/2}(S^m)} \pi(\mathbf{X}_{1:K}^m|\mathbf{Z}_{1:K}(S^m)), \quad (12)$$

and the detection problem (8) can be formulated as

$$\begin{cases} \prod_{m=1}^M \pi(\hat{\mathbf{X}}_{1:K}^m|\mathbf{Z}_{1:K}(S^m)) > V_{DT} & M \text{ targets are declared} \\ \prod_{m=1}^M \pi(\hat{\mathbf{X}}_{1:K}^m|\mathbf{Z}_{1:K}(S^m)) \leq V_{DT} & \text{null is declared.} \end{cases} \quad (13)$$

Whenever a detection is declared, the estimated multi-target state is the concatenation of the estimated individual states.

Intuitively, (12) can be implemented by carrying out a single-target DP integration on the entire measurement sequence $\mathbf{Z}_{1:K}$ and extracting the target state with the maximum merit in each subset of the discretized state space. However, usually both the number of targets present and their initial positions are unknown, that means these subsets can not be pre-defined. Here, the IAT and SAM processes are designed to approximately implement (12)–(13) without prior target information. Before proceeding, it is necessary to introduce the following Definitions and Propositions which are instrumental in the development of IAT and SAM.

For a scenario with M targets, let $A^m = [a_1^m, \dots, a_K^m]$, $m = 1, \dots, M$ denotes the cells containing the trajectory of the m th

target over the K frames, where its k th element $a_k^m \in \overline{\mathbb{R}^2}$. Let $\tilde{\mathbf{Z}}_{1:K}^m$ denote the measurement sequence made up of three parts: the measurements recorded in the cells A^m occupied by the m th target; the zero-value measurements in cells occupied by other targets but not the m th target and the noise measurements for the remaining cells.² Recall that $I(\mathbf{x}_K|\mathbf{Z}_{1:K})$ denotes the K th stage merit function of state $\mathbf{x}_K$ after implementing the single-target DP integration on measurements $\mathbf{Z}_{1:K}$.

Definition 1: The m th target is said to be *detectable* if

$$\mathbf{x}_K^m = \arg \max_{\mathbf{x}_K \in \overline{\mathbb{R}^s}} I(\mathbf{x}_K|\tilde{\mathbf{Z}}_{1:K}^m), \quad (14)$$

$$I(\mathbf{x}_K^m|\tilde{\mathbf{Z}}_{1:K}^m) > V_{DT}. \quad (15)$$

Equation (14) means that $\mathbf{x}_K^m$ is the state which has the highest integration merit after implementation of single target DP on measurements $\tilde{\mathbf{Z}}_{1:K}^m$. Equation (15) further requires that the merit of $\mathbf{x}_K^m$ needs to exceed the detection threshold V_{DT} , not only the highest one. Therefore, a target is *detectable* only if the integrated merit function of its *discretized* true target state is the highest one and also exceeds V_{DT} .

Remark: If the i th target trajectory doesn't share measurement cells with any other targets, i.e., $a_k^i \neq a_k^j$ for $k = 1, \dots, K$, and $i \neq j$, $i, j \in \{1, 2, \dots, M\}$, then the sequence $\tilde{\mathbf{Z}}_{1:K}^i$ is only made up of the contributions of the i th target and the background noise. However, target trajectories may share common cells when targets are in proximity. For example, if $a_k^i = a_k^j$, then both $\tilde{\mathbf{Z}}_{1:K}^i$ and $\tilde{\mathbf{Z}}_{1:K}^j$ contain contributions from the i th and j th targets, and the background noise.

Definition 2: For a scenario with M targets, if the initial range separation $R_{i,j}$ between any two targets i and j is larger than $2R_{\max}$, where $i \neq j$, $i, j \in \{1, 2, \dots, M\}$, then the M targets are *mutually isolated*.

Proposition 1: For a scenario with M targets, if all M targets in a scenario are *mutually isolated*, then, we have

$$I(\mathbf{x}_K^m|\mathbf{Z}_{1:K}) = I(\mathbf{x}_K^m|\tilde{\mathbf{Z}}_{1:K}^m), \quad m = 1, \dots, M. \quad (16)$$

Proof: During the DP implementation, for the possible trajectories of $\mathbf{x}_K^m$ which are included in the discretized state space $\overline{\mathbb{R}^s}(S^m)$, the integration merit of $\mathbf{x}_K^m$ is completely decided by measurements taken in $\overline{\mathbb{R}^s}(S^m)$, i.e., $\mathbf{Z}_{1:K}(S^m)$. Thus

$$I(\mathbf{x}_K^m|\mathbf{Z}_{1:K}) = I(\mathbf{x}_K^m|\mathbf{Z}_{1:K}(S^m)). \quad (17)$$

Because all targets are *mutually isolated*, measurement $\mathbf{Z}_{1:K}(S^m)$ only contains the contribution of noise background and the m th target, we have

$$I(\mathbf{x}_K^m|\mathbf{Z}_{1:K}(S^m)) = I(\mathbf{x}_K^m|\tilde{\mathbf{Z}}_{1:K}^m). \quad (18)$$

For the trajectories of $\mathbf{x}_K^m$ which are out of the scope of $\overline{\mathbb{R}^s}(S^m)$, the merit integration of $\mathbf{x}_K^m$ is also decided by the measurements from background noise and the m th target due to the speed limit, since all targets are mutually isolated. So Proposition 1 is proved. ■

²Note that $\tilde{\mathbf{Z}}_{1:K}^m$ consists both noise background and the m th target, thus represents a sequence of complete measurement frames of size $N_x \times N_y$.

Proposition 2: For a scenario with M targets, if all the M targets in a scenario are *mutually isolated* and *meantime detectable*, then for $m = 1, \dots, M$ we have

$$\begin{aligned} \mathbf{x}_K^m &= \arg \max_{\mathbf{x}_K \in \overline{\mathbb{R}^s}(S^m)} I(\mathbf{x}_K|\mathbf{Z}_{1:K}) \\ &= \arg \max_{\mathbf{x}_K \in \overline{\mathbb{R}^s}} I(\mathbf{x}_K|\tilde{\mathbf{Z}}_{1:K}^m), \end{aligned} \quad (19)$$

$$I(\mathbf{x}_K^m|\mathbf{Z}_{1:K}) = I(\mathbf{x}_K^m|\tilde{\mathbf{Z}}_{1:K}^m) > V_{DT}. \quad (20)$$

Proof: Using Definition 1 and Proposition 1 we have

$$\mathbf{x}_K^m = \arg \max_{\mathbf{x}_K \in \overline{\mathbb{R}^s}} I(\mathbf{x}_K|\tilde{\mathbf{Z}}_{1:K}^m), \quad (21)$$

$$I(\mathbf{x}_K^m|\mathbf{Z}_{1:K}) = I(\mathbf{x}_K^m|\tilde{\mathbf{Z}}_{1:K}^m) > V_{DT}. \quad (22)$$

During the DP implementation, the merit integration of any state $\mathbf{x}_K^a \in \overline{\mathbb{R}^s}(S^m)$ is decided by the measurements taken by $\overline{\mathbb{R}^s}(S^m)$, and sometimes also the measurements from other parts of the measurement frame when $\mathbf{x}_K^a$ is near the boundary of the subset $\overline{\mathbb{R}^s}(S^m)$. However, since all targets are *mutually isolated*, no measurement from other targets can contribute to the merit integration of $\mathbf{x}_K^a$ due to the speed limit. Thus the merit integration of $\mathbf{x}_K^a$ is decided by the measurements from background noise and the m th target. Then

$$\arg \max_{\mathbf{x}_K \in \overline{\mathbb{R}^s}(S^m)} I(\mathbf{x}_K|\mathbf{Z}_{1:K}) = \arg \max_{\mathbf{x}_K \in \overline{\mathbb{R}^s}(S^m)} I(\mathbf{x}_K|\tilde{\mathbf{Z}}_{1:K}^m). \quad (23)$$

Since $\mathbf{x}_K^m \in \overline{\mathbb{R}^s}(S^m)$, from (21), we have

$$\mathbf{x}_K^m = \arg \max_{\mathbf{x}_K \in \overline{\mathbb{R}^s}(S^m)} I(\mathbf{x}_K|\mathbf{Z}_{1:K}). \quad (24)$$

Using (21), (22) and (24) Proposition 2 is proved. ■

Let $\mathcal{D}$ denote the collection of target states whose merit functions exceed V_{DT} after implementing the single target DP integration on the entire measurements $\mathbf{Z}_{1:K}$,

$$\mathcal{D} = \{\mathbf{x}_K : I(\mathbf{x}_K|\mathbf{Z}_{1:K}) > V_{DT}, \mathbf{x}_K \in \overline{\mathbb{R}^s}\}. \quad (25)$$

Using Proposition 2, we have that the merit of all the *detectable* and *mutually isolated* targets can exceed the detection threshold after implementing the single target DP integration on measurements $\mathbf{Z}_{1:K}$. The collection $\mathcal{D}$ contains all the individual target states, and meanwhile a lot of redundant state estimates caused by the merit function expansion phenomenon. Besides, the states of these *detectable* and *mutually isolated* targets are the local merit maxima in each of the subsets of the discretized state space $\overline{\mathbb{R}^s}(S^1), \dots, \overline{\mathbb{R}^s}(S^M)$. This suggests that an alternative way to implement (12) and (13) is first forming the collection $\mathcal{D}$ (i.e., IAT), then extracting the individual target states from $\mathcal{D}$ (i.e., SAM). The pseudo code of IAT is shown in Table I. As the name suggests, IAT first implements a single target DP integration, then applies a threshold on the final stage merit functions to form $\mathcal{D}$.

The number of state candidates in $\mathcal{D}$ is potentially very large. Fig. 4 shows the recovered trajectories of the three-target scenario in Fig. 1. It can be seen that there are three separated target

TABLE I
THE PSEUDO CODE OF INTEGRATION AND THRESHOLDING PROCESS

- DP integration: For $1 \leq k \leq K$ and for all $\mathbf{x}_k \in \mathbb{R}^s$

$$I(\mathbf{x}_k | \mathbf{Z}_{1:k}) = \mathcal{F}(\mathbf{z}^k, I(\mathbf{x}_{k-1} | \mathbf{Z}_{1:k-1})) \quad (26)$$
 where function $\mathcal{F}$ means the single target DP recursion.
- Form the candidate collection: For $k = K$, find

$$\mathcal{D} = \{\mathbf{x}_K : I(\mathbf{x}_K | \mathbf{Z}_{1:K}) > V_{DT}, \mathbf{x}_K \in \mathbb{R}^s\} \quad (27)$$

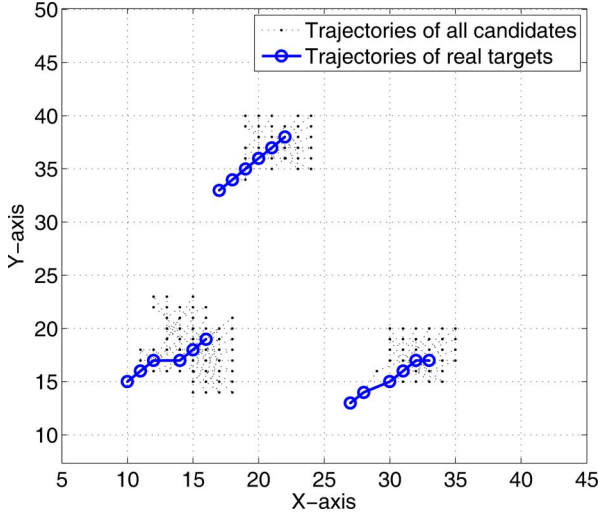


Fig. 4. The recovered trajectories of the three-target scenario in Fig. 1 after IAT. The real target trajectories are plotted by circled-line and the candidate trajectories are plotted by dash-dot line.

clusters, each one corresponds to a target. Most of the recovered trajectories are branches of the real target trajectories due to merit function expansion. To delete the inappropriate candidates, the SAM process evaluates all the elements in $\mathcal{D}$ for both the quality of merit function and uniqueness of history over the observation period. The pseudo code of SAM is shown in Table II. Firstly, SAM finds the trajectory which has the maximum merit function in $\mathcal{D}$. We refer to this as the original trajectory. All trajectories which are similar [6], [17], [26], in some sense, to the original-trajectory are selected, and these trajectories are referred to as sub-trajectories. Since all the sub-trajectories are likely to have originated from the same target as the original-trajectory, all sub-trajectories are deleted and only the original-trajectory is added to the collection of declared targets $\mathcal{D}^*$. If there are still trajectories in $\mathcal{D}$ (multi-target scenario), then find the next original-trajectory and delete its corresponding sub-trajectories. After the recursive processing of SAM, the redundant candidates are removed from $\mathcal{D}$, all the declared target estimates are stored in $\mathcal{D}^*$ and the estimated number of targets is the number of elements in $\mathcal{D}^*$.

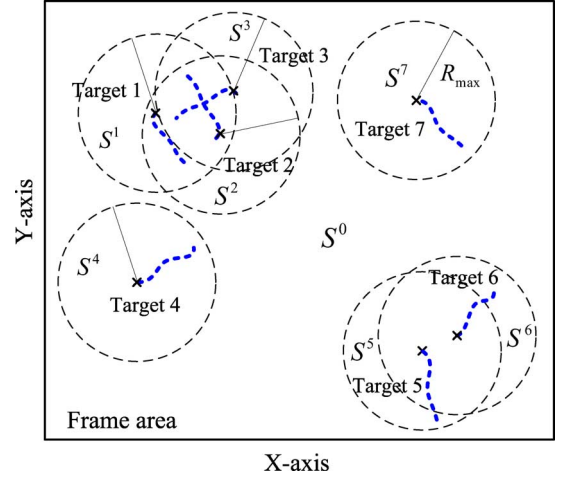
There are many ways to judge the similarity between two trajectories. One commonly used method is to select the trajectories which share at least L common states with this original-trajectory as the sub-trajectories [6]. This strategy works well when targets are well separated. However, when targets are in proximity, situations become more complicated, and stricter ways of cutting branches of the declared targets may be required. For instance, all the branches whose distance to the original-trajectory are smaller than a threshold V_c , rather than sharing at least

 TABLE II
THE PSEUDO CODE OF SORT AND MERGE PROCESS

- 1) Find out the m th original-trajectory $\mathbf{y}_{1:K}^m$ as:

$$\mathbf{y}_K^m = \arg \max_{\mathbf{x}_K \in \mathcal{D}} I(\mathbf{x}_K) \quad (28)$$
- 2) Find out the corresponding sub-trajectories of $\mathbf{y}_{1:K}^m$ which share the similarity with $\mathbf{y}_{1:K}^m$.
- 3) Merge the trajectory-group: remove $\mathbf{y}_{1:K}^m$ and its corresponding sub-trajectories from $\mathcal{D}$.
- 4) Update the collection $\mathcal{D}^*$ of the declared targets:

$$\mathcal{D}^* = \{\mathbf{y}_K^1, \dots, \mathbf{y}_K^m\} \quad (29)$$
- 5) Recursion: if $\mathcal{D} \neq \emptyset$, repeat the step 1)-4).


 Fig. 5. Sketch map of a scenario wherein 7 targets are arbitrarily located. The dash circles indicate the circular areas $S^1, \dots, S^M$ containing the target trajectory. The remaining part of the scenario S^0 only contains the noise background.

L commons cells with the original-trajectory, are selected as sub-trajectories. Otherwise, some branches which share one or two targets measurements may survive and be declared as targets when SNR is high, resulting in a high rate of false alarm. This is the approach used here.

B. Tracking Arbitrarily Located Targets

The assumption that all targets are spaced far apart is usually too stringent for realistic problems. The ability of a tracking system to deal with difficult conditions such as target crossing is much valued in practical applications. Here, the MTT is considered for the more general cases where M targets are located arbitrarily. The M target trajectories can be included in M circular areas $S^1, \dots, S^M$. However, since the location of targets are arbitrary, some circular areas may intersect with each other (i.e., the corresponding discretized state spaces of these areas share common states). Consider that M targets are separated into $0 < l \leq M$ clusters such that targets whose state subsets intersect with each other belong to the same cluster. Let $C^i, i = 1, \dots, l$ denote the collection of target indices of the i th cluster with $|C^i|$ the number of elements in this collection. Fig. 5 is a sketch map of a scenario with 7 targets. It can be seen that there are $l = 4$ clusters, e.g., targets 1, 2 and 3 make up the first cluster with target number $|C^1| = 3$, and the second cluster has only target 4, $|C^2| = 1$.

For this scenario, the posterior density of the multi-target state can be factorized as in (10). Consequently, for those clusters which have only a single target, the state estimation can be treated as a ST-DP-TBD problem and handled by IAT and SAM; while for those clusters which contain more than one target, the estimation problem should be considered jointly and implemented by solving the following joint maximization

$$\begin{aligned} \hat{\mathbf{C}}_{1:K}^i &= \max_{\mathbf{C}_{1:K}^i \in \overline{\mathbb{R}^{s^K}}^{|C^i|}(V^i)} p(\mathbf{C}_{1:K}^i | \mathbf{Z}_{1:K}(V^i)), \\ \text{s.t. } p(\hat{\mathbf{C}}_{1:K}^i | \mathbf{Z}_{1:K}(V^i)) &> V_{DT}, \end{aligned} \quad (30)$$

with V^i the frame area which is taken by the $|C^i|$ intersecting circular areas of the i th cluster

$$V^i = \bigcup_{j=1}^{|C^i|} S^{C^i(j)}, \quad i = 1, \dots, l, \quad (31)$$

where $C^i(j)$ is the index of the j th target in the i th cluster.

In order to solve the joint maximization in (30), one possible way is to use the K -path DP algorithms [27]–[29]. Although K -path DP can provide an efficient way to implement (30), its complexity still increases exponentially in the number $|C^i|$ of targets and linear in the number K of processed frames, which is not viable for practical applications. Moreover, a multi-hypotheses detection algorithm is again required, since the dimension of $\mathbf{C}_{1:K}^i$ has to be determined before implementing K -path DP. Even if we would like to do this, there is another problem: the prior target information required to pre-define the number of state subsets is usually unavailable. Therefore, we propose the PTC-DP process to sub-optimally implement (30) in a fast manner without the prior target information. Before introducing PTC-DP, Proposition 3 needs to be given first.

Proposition 3: Assume a scenario with M targets separated into $0 < l \leq M$ clusters such that targets whose state subsets intersect with each other belong to the same cluster. If the following two assumptions hold,

- all the M targets are *detectable*
- there are M_d , ($1 \leq M_d < M$) valid detections declared after the implementation of IAT and SAM on $\mathbf{Z}_{1:K}$.

Then

- each of the target clusters has at least one target being detected, that means $M_d \geq l$.
- the detection misses of the $M - M_d$ *detectable* targets are because of the interference from the M_d declared targets. The initial positions between the missed targets and interference targets are less than $2R_{\max}$.

Proof: For the special cases where all the M targets are spaced far apart such that all targets are *mutually isolated*, targets should be clustered into $l = M$ groups. Using Proposition 2, we have $M_d = l$ and $M - M_d = 0$.

For closely spaced targets, a target cluster may contain more than one target. Let $\mathbf{X}_{1:K}^e = (\mathbf{x}_1^e, \dots, \mathbf{x}_K^e)$ and $A^e = (a_1^e, \dots, a_K^e)$ denote a state sequence in a certain target cluster and the cells occupied by its trajectory. If this cluster contains other targets in addition to target e , there are two possible cases. In the first case, target e shares at least L common states with a declared target. Then the state sequence of target

e is one of the sub-trajectories of that declared target, and will be deleted during the processing of SAM. Note that the initial positions between the missed targets and declared targets must be less than $2R_{\max}$, otherwise, they can't share common states because of the speed limit. The second case is that the state sequence of target e shares less than L common states with any declared targets. Thus based on assumption one, target e will be declared after IAT and SAM. Therefore, for any target cluster, at least the strongest target will be declared after IAT and SAM. That means $M_d \geq l$. So Proposition 3 is proved. ■

Proposition 3 indicates that if there are M_d targets declared after the IAT and SAM process, the potentially missed targets must be around these declared targets. Hence, there is no need to characterize each of the discretized state subsets in advance. We only need to explore the vicinities $L^1, \dots, L^{M_d}$ of the M_d declared targets. Now, the joint-maximization (30) can be implemented without the characterization of measurement circles, but still a multi-hypotheses detection algorithm is required because of the unknown number of the missed targets in these vicinities. To avoid the complexity of high-dimensional maximization and the multi-hypotheses testing problem, a suboptimal solution called target-cancellation DP (TC-DP) is designed here to further split the joint maximization for each target cluster into single ST-DP-TBD problems.

The pseudo code of TC-DP is shown in Table III. The term $\hat{\mathbf{Z}}_{1:K}(L^{i,j})$ in (32) is defined as the measurement vector in the region L^i with the interference of the first $j - 1$ declared targets removed. When $j = 1$, $\hat{\mathbf{Z}}_{1:K}(L^{i,1}) = \mathbf{Z}_{1:K}(L^i)$. There are a number of ways of removing the interference of declared targets. Perhaps the best way is to account for the declared targets in the likelihood evaluation. While this approach is likely to provide good performance, it raises difficulties in the setting of the threshold. We therefore adopt a simpler approach in which the measurements in cells occupied by declared targets are set to zero³. With this method targets which are yet to be declared are effectively barred from cells occupied by declared targets. Thus, in cases where a yet to be declared target occupies the same cell as a declared target, it is forced to select a neighboring cell. While this reduces the ability to detect closely spaced targets it will be seen in Section VI that good performance can still be achieved provided a sufficient number of strong measurements from unoccupied cells are available. Another simple but useful way to alleviate target information loss is to fill the cleared cells with randomly generated measurements based on the distribution of noise, as there would still be a chance that the cell would be selected.

Through the recursive solution in Table III, prospective targets around the i th detected target can be extracted one-by-one from the vicinity of state space, i.e., $\overline{\mathbb{R}^{s^K}}(L^i)$. Note that, there is no need to pre-set the number of iteration times for each TC-DP, the recursion of TC-DP stops when there are no merit function values exceeding V_{DT} after the second step of TC-DP. Hence, if there are M_d targets declared after the IAT and SAM process, M_d TC-DP algorithms are implemented simultaneously in the

³The idea of reprocessing data frames after the target related measurements has been removed was mentioned before in [5]. Also it is similar in spirit to the CLEAN [32], [33] algorithm in multiple-element interferometry in radio astronomy, which successively removes large targets and their sidelobe responses by subtracting the point spread function at the locations of the strong targets.

TABLE III
 THE PSEUDO CODE OF TARGET-CANCELLATION DP PROCESS

 For the vicinity of the i th declared target, the TC-DP proceeds as,

- 1) Initialization: $j = 1$, $\hat{\mathbf{Z}}_{1:K}^j(L^{i,1}) = \mathbf{Z}_{1:K}(L^i)$.
- 2) Extract the strongest target:

$$\hat{\mathbf{X}}_{1:K}^j = \arg \max_{\mathbf{x}_{1:K}^j \in \mathbb{R}^{sK}(L^i)} p(\mathbf{x}_{1:K}^j | \hat{\mathbf{Z}}_{1:K}^j(L^{i,j})), \quad (32)$$
- 3) IF $p(\hat{\mathbf{X}}_{1:K}^j | \hat{\mathbf{Z}}_{1:K}^j(L^{i,j})) > V_{DT}$,
 - Add the state of the j th extracted target $\hat{\mathbf{x}}_K^j$ in the declared target collection $\mathcal{D}^*$,
 - Remove the interference of the extracted target,
 - Go to step two and $j = j + 1$.
- 4) ELSE
 - End recursion

M_d corresponding vicinities. We refer to this process as the parallel TC-DP (PTC-DP). PTC-DP allows possibly missed targets to be extracted group-by-group by repeatedly searching the potential target-hidden areas. The size of the potential target-hidden areas should be carefully chosen, because it balances the conflicts between computational expense and detection performance. The appropriate selection of these areas will be discussed in the next subsection.

C. Selecting the Size of the Potential Target-Hidden Areas

To guarantee that all the possibly missed targets (after IAT and SAM) can be covered by the search of PTC-DP, the potential target-hidden areas, $L^1, \dots, L^{M_d}$ should be set large enough. Obviously, the safest way to select L^i , $i = 1, \dots, M_d$, is to cover the frame area with radius $R_{\max}$, centered at the initial position of the i th detected target. In this case, the number of discretized states in its state subset $\mathbb{R}^{sK}(L^i)$ is,

$$|\mathbb{R}^{sK}(L^i)| = f(R_{\max})(2V_x^{\max} + 1)(2V_y^{\max} + 1)K, \quad (33)$$

where $f(R)$ is the number of frame cells covered by a circle with a radius R . Equation (33) is an upper bound on the size of $\mathbb{R}^{sK}(L^i)$, but we don't have to select $\mathbb{R}^{sK}(L^i)$ as shown. To keep the computational expense down, the size of the potential target-hidden areas should be set as small as possible without affecting target detection performance.

Here, we consider the size selection of $L^1, \dots, L^{M_d}$ by finding out the minimum position distance in which target interference can affect the tracking performance. As shown in Fig. 6(a), two scenarios are considered: two parallel targets and two following targets. The interference intensity between targets is reflected by their mutual distances, i.e., parallel distance d_p and following distance d_f . The maximum target velocities are set as $V^{\max} = 2$ cell/frame time for both dimensions, and the number of integrated frames is $K = 6$. The results are gathered by averaging 100 realizations.

Fig. 6(b) shows the number of valid detected targets of the two scenarios. When SNR = 12 and 15 dB, IAT and SAM can successfully detect both targets if the target distance is larger than 3 cells. However, when the target distance is below 3 cells, the detection performance suffers considerably from target interference. Similar phenomenon can also be observed

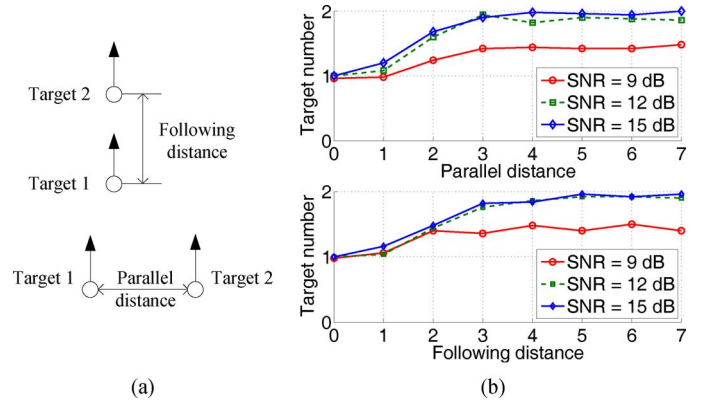


Fig. 6. (a) Two target scenarios for examining the effect of target interference on tracking performance. (b) The number of valid detected targets is plotted against several different target distances for SNR = 9, 12, and 15 dB.

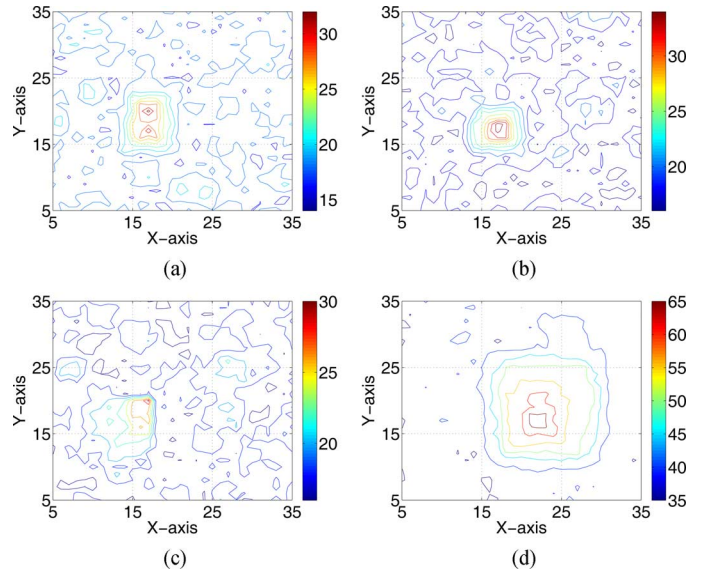


Fig. 7. The contour maps of the single-target DP merit function for two parallel targets with different integration length, maximum velocity and parallel distances. (a) $K = 6$, $d_p = 3$, $V^{\max} = 2$. (b) $K = 6$, $d_p = 1$, $V^{\max} = 2$. (c) $K = 6$, $d_p = 3$, $V^{\max} = 3$. (d) $K = 12$, $d_p = 3$, $V^{\max} = 2$.

for SNR = 9 dB, although the overall detection performance degrades because of the low amplitudes of target signals. The results indicate that, for this simulation setup, a possible choice for the setting of the potential target-hidden areas is 3 cells around the trajectory of detected targets.

Generally for larger values of $V^{\max}$ and K , the expected size of $L^1, \dots, L^{M_d}$ should be larger. This is understandable based on (33) which indicates that the upper bound size of the state space of the potential target-hidden area increases as $V^{\max}$ and K increases. Fig. 7 shows the contour maps of the single-target DP merit function for two parallel targets with different values of K , $V_{\max}$ and d_p . It can be seen from Fig. 7(a) that the merit peaks of the two parallel targets are separated when $K = 6$, $d_p = 3$ cells and $V^{\max} = 2$ cell/frame time, but the peaks are merged together when d_p reduces to 1 cell (see Fig. 7(b)). This is consistent with the results in Fig. 6(b). Fig. 7(c) and (d) show that if we increase either K or $V^{\max}$, the target peaks will merge together even for $d_p = 3$ cells. This is because the merit function expansion phenomenon becomes severer as K and $V_{\max}$ increase so that a larger size of $L^1, \dots, L^{M_d}$ is required.

D. Computational Expense

The bulk of the computational expense of MT-DP-TBD is spent on two kinds of DP enumerations: the single-target DP integration in the IAT process with a computational cost $\mathcal{O}(K|\mathbb{R}^s||\tau|)$, and the TC-DP recursions in the PTC-DP process. Consider that M_d targets are declared after IAT and SAM, then M_d TC-DP algorithms are implemented simultaneously. The computational cost of each TC-DP enumeration is about $\mathcal{O}(K|\mathbb{R}^s(L^i)||\tau|)$, where $|\mathbb{R}^s(L^i)|$ is calculated as

$$|\mathbb{R}^s(L^i)| = (2\hat{R}_x + 1)(2\hat{R}_y + 1)(2V_x^{\max} + 1)(2V_y^{\max} + 1), \quad (34)$$

with $\hat{R}_x, \hat{R}_y$ the minimum target interference distances described in Section V-C. Usually $\hat{R}_x$ and $\hat{R}_y$ are much smaller than the length of x and y dimensions, which indicates the computational expense of TC-DP is much lower than the single target DP integration on the entire measurement frame. The complexity of a TC-DP enumeration compared to the DP enumeration in IAT is

$$\frac{\mathcal{O}(K|\mathbb{R}^s(L^i)||\tau|)}{\mathcal{O}(K|\mathbb{R}^s||\tau|)} = \frac{(2\hat{R}_x + 1)(2\hat{R}_y + 1)}{N_x N_y}. \quad (35)$$

For example, if the observation area is an $x - y$ plane with $N_x = N_y = 100$, and $\hat{R}_x = \hat{R}_y = 3$. Then the computation cost of each TC-DP iterative run is only about 0.5% of the DP enumeration in IAT. Hence, even if several groups of targets are in close proximity in the surveillance area, the computational cost of performing PTC-DP is negligible compared to the cost of the initial DP enumeration. Besides, even if the safest choice of L^i is considered, i.e., (33), the complexity of a TC-DP enumeration compared to the DP enumeration in IAT is

$$\frac{f(R_{\max})}{N_x N_y} < \frac{4R_{\max}^2}{N_x N_y}. \quad (36)$$

Consider a maximum velocity $V^{\max} = 2$ cell/frame for each direction, and a integration length $K = 6$, then $R_{\max} = 12\sqrt{2}$ and the computation cost of each TC-DP iterative run is less than 11.5% of the DP enumeration in IAT. The computational efficiency of TC-DP will become more conspicuous as the size of surveillance area increases.

E. Discussion

A brief summary of the proposed MT-DP-TBD algorithm is given in Table IV. It is worth mentioning that both Propositions 2 and 3 are based on the assumption that the targets are *detectable*. This implies that if a target is not *detectable* (unable to be detected by ST-DP-TBD when only this target exists), it will not be detected by MT-DP-TBD. For this reason, the performance of MT-DP-TBD is generally upper bounded by the performance of ST-DP-TBD. Note that, if the set $\mathcal{D}$ of candidate states obtained in the initial single-target DP is empty then $M = 0$ targets are assumed to be present.

In essence, MT-DP-TBD is an approximation to the optimal multi-target state based DP-TBD. Compared to the sub-optimal algorithms in [26], the approximation of MT-DP-TBD is more accurate and computationally efficient. SP-STC-VTA [26] involves only a single DP search, and also a type of SAM since it removes from consideration all branches from detected

TABLE IV
PROPOSED MT-DP-TBD ALGORITHM FOR THE DETECTION
AND TRACKING OF MULTIPLE TARGETS

Integration and thresholding process
• Implement the single-target DP integration using (7). • Threshold the DP merit function, then form the collection $\mathcal{D}$ of the candidate states group using (27).
Sorting and merging process
• Form the trajectory-group using (28). • Merge the trajectory-group using (29). • Recursion. • Build up the collection $\mathcal{D}^*$ of the declared targets.
Parallel-target-cancellation DP process
• Extract the strongest target using (32). • Remove the target interference of the previously detected targets. • Recursion. • Add the newly extracted targets to collection $\mathcal{D}^*$ of the declared targets.

targets. Its complexity is similar to MT-DP-TBD, but as will be shown in the simulation results, it provides comparatively poor accuracy and decreased detection for closely-spaced targets. STC-VTA [26] involves repeated searches after removing previously detected trajectories, which is similar to MT-DP-TBD. However, it does not modify the measurements to remove the contributions of previously detected targets, as we do in our algorithm. This is an important difference as the merit function expansion noted in Section IV suggests that merely removing a path from consideration will not prevent many nearby paths from also being reported as detected⁴. STC-VTA is also much more computationally expensive because its re-searching process includes the whole state space rather than just a neighborhood, as given in (34). Further, it adopts a nested multiple hypothesis testing detection procedure so that, even if no target is present, $M_{\max}$ searches over the state space are performed. In contrast, the proposed MT-DP-TBD makes no assumption on the maximum number of targets and, since sequential track extraction continues only as long as the measurements suggest, redundant computations are avoided.

VI. NUMERICAL EXAMPLE

In this section, the performances of the previously proposed SP-STC-VTA, STC-VTA [26] and MT-DP-TBD are investigated using the following performance measurements:

- 1) *The probability of target detection P_d* : the probability that the last stage positions of the estimated state are within 2 cells of the actual target cell, as in [6], [7].
- 2) *The probability of false alarm P_{fa}* : the probability of reporting at least one false track when no target appears. This is equal to the probability of the maximum merit of noise state exceeding the threshold V_{DT} after implementing ST-DP-TBD.
- 3) *The number of the valid detections N_{vd}* : the average number of the declared targets whose last stage positions are within 2 cells of the actual target cell.

⁴For the 2-D position only state-space considered in [26], removing a path from consideration has the same effect as modifying the measurements to remove the contributions of previously detected targets. However, when the dimension of the state space is higher than that of measurement space, suppressing a particular path in the state space is not enough to remove the influence of previously detected targets.

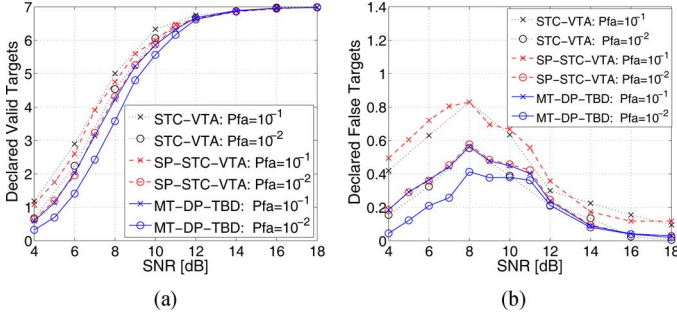


Fig. 8. The numbers N_{vd} of valid detections and the numbers N_{fd} of declared false targets are plotted against SNR from 4 dB to 18 dB for STC-VTA, SP-STC-VTA and MT-DP-TBD. (a) N_{vd} . (b) N_{fd} .

- 4) The number of the declared false tracks N_{fd} : the average difference between the number of declared targets and the number of valid detections.
- 5) The root mean square (RMS) position error of the valid tracks: the average position difference between the estimated trajectory of the valid target detection and the discretized trajectory of the real target.

The performance of SP-STC-VTA, STC-VTA and the proposed MT-DP-TBD is first assessed in a scenario with seven well-separated targets. The detection thresholds for all methods are chosen to achieve a certain false alarm probability. The number of integrated frames is $K = 6$, and $V_x^{\max} = V_y^{\max} = 3$ cell/frame. The upper bound of the number of the potential targets for SP-STC-VTA and STC-VTA is chosen as $M_{\max} = 10$. For this scenario, we can simply use the branch cutting criteria which only removes branches with common cells of detected targets. In the following analysis, the results are gathered by averaging over 500 realizations.

In Fig. 8(a), the number N_{vd} of valid detections is plotted against SNR for $P_{fa} = 10^{-1}$ and 10^{-2} . It can be seen that all three methods perform well, and can correctly detect all targets when SNR is high. Noticeably, the performance of SP-STC-VTA and STC-VTA is even better than MT-DP-TBD. This is because there is a bias in the threshold setting of SP-STC-VTA and STC-VTA [26], i.e., the false alarm probability of overestimating targets is not included in the calculation of P_{fa} . Therefore, the selected detection threshold for SP-STC-VTA and STC-VTA is lower than it should be. This also explains that why the numbers N_{fd} of false targets declared by SP-STC-VTA and STC-VTA are greater, which can be seen clearly in Fig. 8(b). In the simulation, the upper bound of the number of the potential targets is set as $M_{\max} = 10$ for SP-STC-VTA and STC-VTA. A higher false alarm rate might be observed for larger values of $M_{\max}$ as a consequence of more severely biased threshold setting.

In Fig. 9, the detection probability P_d and RMS position error of all seven targets are plotted against SNR for $P_{fa} = 10^{-1}$. The P_d of ST-DP-TBD and single-pulse detector [34], and the RMS position error of ST-DP-TBD are also given as benchmarks. As expected, the detection results of all targets approach to 1 as the target SNR increases. Note that unlike the P_d of MT-DP-TBD which is upper bounded by the P_d of ST-DP-TBD, the P_d of SP-STC-VTA and STC-VTA is

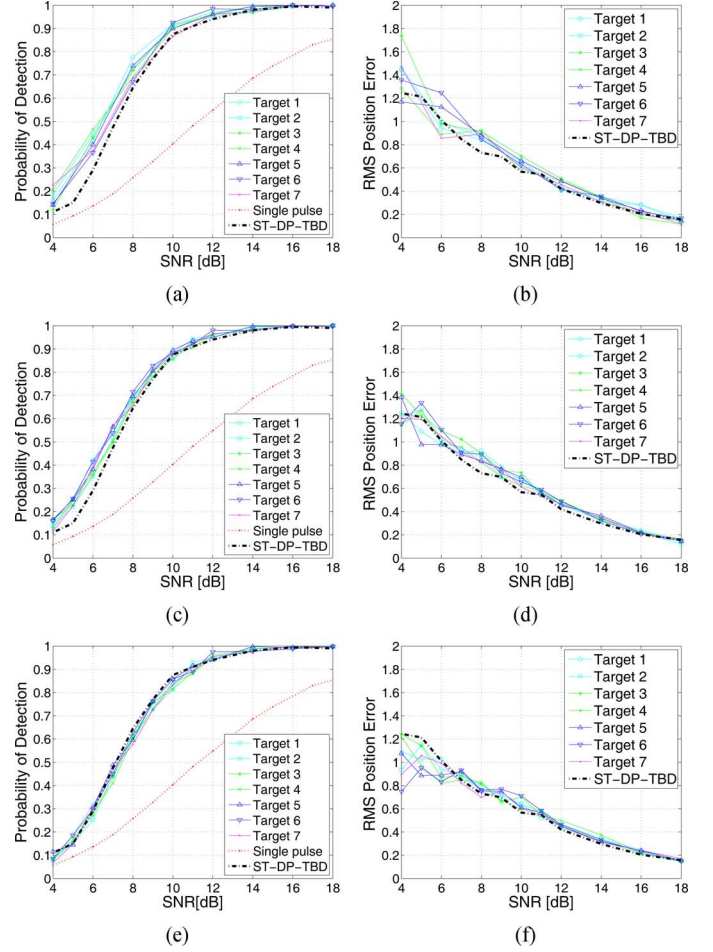


Fig. 9. The detection probability P_d and RMS position errors of all targets are plotted against SNR from 4 dB to 18 dB for $P_{fa} = 10^{-1}$. The P_d of ST-DP-TBD and single-pulse detector, and the RMS position error of ST-DP-TBD are also given as benchmarks. (a) P_d of STC-VTA. (b) RMS position error of STC-VTA. (c) P_d of SP-STC-VTA. (d) RMS position error of SP-STC-VTA. (e) P_d of MT-DP-TBD. (f) RMS position error of MT-DP-TBD.

slightly above the P_d of ST-DP-TBD due to their biased thresholds. But in general, the results of Figs. 8 and 9 indicate that SP-STC-VTA, STC-VTA and MT-DP-TBD perform equally well when no target interference exists. In regard to tracking accuracy, it can be seen that the RMS position errors of all targets for SP-STC-VTA, STC-VTA and MT-DP-TBD are similar to that of ST-DP-TBD.

Next, a more difficult scenario, as shown in Fig. 10, is designed to assess the performance of three methods in a complicated multi-target environment. Seven targets, which may be separated into 4 clusters, move in a 50×50 observation area. Target 7 is an isolated target located far from all other targets. Targets 1 and 2 are close to each other and moving in parallel, but never occupy the same grid cell. Targets 3 and 4, targets 5 and 6 are two pairs of crossing targets. For this scenario, stricter branch cutting criteria which removes branches within a certain distance of detected targets is required to avoid false declarations due to the target interference.

Fig. 11(a) reports the numbers N_{vd} of valid detections for all three methods. It can be seen that SP-STC-VTA performs worst due to the target interference when targets are in proximity. It basically provides one detection for each target cluster.

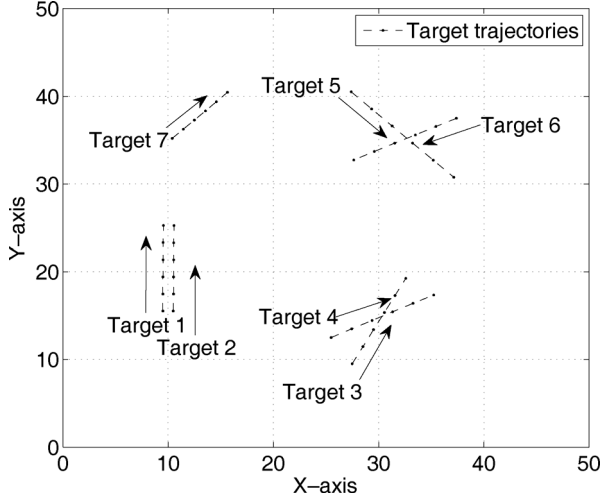


Fig. 10. The schematic showing of the motion of the seven targets in the simulation scenario. The target motions are indicated by the arrows.

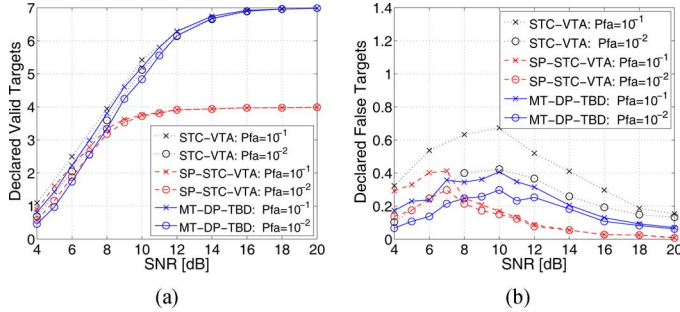


Fig. 11. The numbers N_{vd} of valid detections and the numbers N_{fd} of declared false targets are plotted against SNR from 4 dB to 20 dB for STC-VTA, SP-STC-VTA and MT-DP-TBD. (a) N_{vd} . (b) N_{fd} .

This happens because SP-STC-VTA only involves a single DP search so that, once a target in a cluster is detected, the tracks of closely spaced targets in the same cluster will be cleared during SAM and cannot be found in later iterations. By comparison, STC-VTA and MT-DP-TBD can consistently detect all seven targets. STC-VTA reports a slightly larger number of valid tracks compared to MT-DP-TBD but, as can be seen in Fig. 11(b), this slight improvement comes at the expense of a much larger number of false tracks. The false track rate of SP-STC-VTA is reduced considerably compared to MT-DP-TBD and STC-VTA. This is because the branch cutting criteria used here removes many more tracks than in the previous scenario. If the branch cutting criteria is changed so that only branches with common cells are removed, rather than those within a certain distance, then the SP-STC-VTA can detect all seven targets. However, this also results in a larger number of false tracks. Thus there appears to be no simple way for the SP-STC-VTA to resolve closely spaced targets. STC-VTA and MT-DP-TBD can detect closely-spaced targets because both methods involve repeated searches after removing the interference of previously detected trajectories, but as discussed in Section V-E, the implementation of MT-DP-TBD is much more efficient than STC-VTA. This can be seen in Table V which shows the average execution times of the MATLAB implementations of the three algorithms. Note

TABLE V
EXECUTION TIMES (IN SECONDS) OF ALL THREE METHODS

Method	First scenario	Second scenario
1. STC-VTA	127.19	126.58
2. SP-STC-VTA	19.61	19.67
3. MT-DP-TBD	23.92	30.95

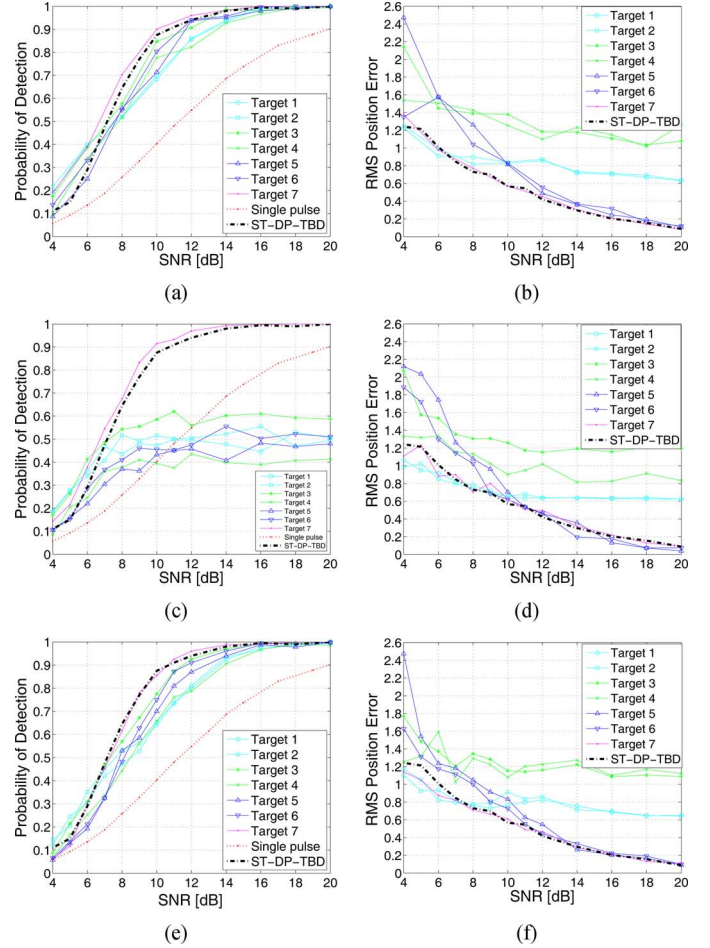


Fig. 12. The detection probability P_d and RMS position errors of all targets are plotted against SNR from 4 dB to 20 dB for $P_{fa} = 10^{-1}$. The P_d of ST-DP-TBD and single pulse detector, and the RMS position error of ST-DP-TBD are also given as benchmarks. (a) P_d of STC-VTA. (b) RMS position error of STC-VTA. (c) P_d of SP-STC-VTA. (d) RMS position error of SP-STC-VTA. (e) P_d of MT-DP-TBD. (f) RMS position error of MT-DP-TBD.

that, unlike SP-STC-VTA and STC-VTA, the running time of MT-DP-TBD depends on the spatial distribution of targets.

Fig. 12 shows the detailed detection probability and RMS position error of all seven targets for all methods with $P_{fa} = 10^{-1}$. It can be seen more clearly in this figure that the poor performance of SP-STC-VTA is due to the low detection probabilities of the closely spaced targets. Besides, the tracking accuracy of targets 1 and 2 and targets 3 and 4 becomes much worse than in the first scenario for all methods due to the strong target interference.

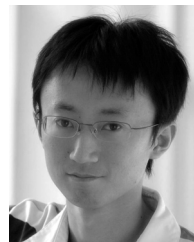
VII. CONCLUSION

In this paper, we consider the tracking of an unknown number of targets through the use of the DP based TBD. To combat the

troublesome high-dimensional optimization problem, the structure of the MTT problem is exploited in the DP implementation so that only adjacent targets are considered jointly, while isolated targets are considered independently. Consequently, the proposed algorithm has much lower computational complexity than implementing DP over the full multi-target state but performs robustly. Moreover, this algorithm adopts a simple and flexible detection procedure to handle the unknown number of targets such that the dimension of the multi-target state no longer needs to be pre-defined before DP enumeration. Both single-target and multi-target scenarios are handled within the same framework without using a multi-hypothesis testing detector. Simulations results show that the proposed algorithm can accurately estimate the number of targets and reliably track multiple targets even when targets are in proximity. During the TC-DP processing, the interference from previously declared targets is removed by simply clearing the measurement in that cell. As we discussed previously, this would lead to considerable performance loss when targets share many common cells. More efficient implementations of target interference cancellation is a topic for further work.

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